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SUPERSEDING
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24 October 1957

MILITARY SPECIFICATION

CANDLES, ILLUMINATING, SURVIVAL, HIGH MELTING POINT

This specification is approved for use by San Antonio ALC, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers high melting point candles of 10-hour duration.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-636	Boxes, Shipping, Fiberboard
PPP-B-676	Boxes, Set Up
PPP-T-60	Tape: Packaging, Waterproof

Military

MIL-P-116	Preservation, Methods Of
MIL-T-22085	Tapes, Pressure-Sensitive Adhesive, Preservation And Sealing

STANDARDS

Military

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-129	Marking For Shipment And Storage
MIL-STD-143	Specifications And Standards, Order Of Precedence For The Selection Of
MIL-STD-831	Test Reports, Preparation Of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: San Antonio Air Logistics Center, Engineering Division, ATTN: MMEDO, Kelly AFB TX 78241, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6260

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, Room 202, Union Station, 516 West Jackson Boulevard, Chicago IL 60606.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

* 3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials.

3.3.1 The manufacturer is permitted wide latitude in the selection of raw materials provided the candles meet the requirements of this specification.

3.3.2 Wick. The wick of the candle shall be of a satisfactory quality braided yarn.

3.4 Performance.

* 3.4.1 Effect of temperature. The candle, when tested as specified in 4.6.2, shall not show that wax flowed after a period of not less than 24 hours, at +125° Fahrenheit (F) ±3°F.

* 3.4.2 Deformation. The candle shall not deform when tested as specified in 4.6.2.

* 3.4.3 Burning quality. The flame shall consume the candle in not less than 10 hours when burned as specified in 4.6.3, without dripping, excessive smoking, or objectionable odor, and without the necessity of trimming the wick.

* 3.5 Weight. The weight of the candle including the can specified in 3.6. shall be not more than 130 grams.

* 3.6 Size. The candle shall be contained in a .012 (100H14) aluminum round slip cover can, nominal size 2-1/4 inches diameter by 2 inches high, body inside lock seam, top edge of body plain, single external bead near top of body for plain edge cover to seat against, bottom flanged, doped double seam with side seam striped with vinyl lacquer. Fill the container with wax to 1/4 inch below top edge. Seamless can with same nominal dimensions will be acceptable.

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* 3.7 Sealing. The can lid shall be sealed with 1 inch pressure sensitive tape that meets the requirements of Type I, MIL-T-22085.

* 3.8 Workmanship. The product shall be prepared by such methods as will reflect good standards of workmanship and quality in the finished product. The wick of the candle shall be properly centered.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of tests. The inspection and testing of candles shall be classified as follows:

a. Quality conformance inspection — See 4.3

b. Preproduction testing — See 4.4

* 4.3 Quality conformance inspection. Quality conformance inspection shall consist of sampling plans and tests.

* 4.3.1 Sampling plans and tests.

* 4.3.1.1 Sampling Plan A. Candles shall be inspected by sampling methods in accordance with MIL-STD-105, for the examination of product test, as described under 4.6.

* 4.3.1.1.1 At the discretion of the inspecting agency, any action specified in MIL-STD-105 for accomplishment by the Government shall be interpreted to mean action to be accomplished either by or under the supervision of the Government, with the final decision as to acceptance or rejection in either case resting with the Government inspector.

* 4.3.1.2 Sampling Plan B.

* 4.3.1.2.1 Lot. For purpose of the sampling plan test, a lot shall consist of all candles which are constructed from the same batch of materials, and submitted for inspection at substantially the same time.

* 4.3.1.2.2 Sampling tests. Eight candles shall be selected at random from each lot and subjected to the tests listed below and described under 4.6.

a. Effect of temperature

b. Burning quality

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* 4.3.1.3 Rejection. When one or more items fail to meet the specified sampling plan tests, acceptance of all items in the lot shall be withheld until the extent and cause of failure are determined. Before resubmitting the lot for a repeat of the sampling plan tests, the contractor shall fully explain to the inspector both the corrections made and the cause of previous rejections.

* 4.4 Preproduction testing.

* 4.4.1 Preproduction test sample tested by the contractor. The contractor shall subject candles to the preproduction tests specified in 4.4.4.

* 4.4.2 Preproduction test report. After the contractor completes the preproduction tests, he shall prepare a preproduction test report in accordance with MIL-STD-831 and furnish a copy of the report to the Government procuring activity.

* 4.4.3 Preproduction test sample for the procuring activity. Along with the preproduction test report, the contractor shall submit one sample to the Government procuring activity.

* 4.4.4 Preproduction tests. Preproduction tests shall consist of all tests described under 4.6.

* 4.5 Test conditions.

* 4.5.1 Atmospheric conditions. Unless otherwise specified, all tests required by this specification shall be made at an atmospheric pressure of 28 to 32 inches of mercury and at a temperature of 77°F, ±18°F, and a relative humidity of 80 percent or less. Where tests are made with atmospheric pressure or temperature substantially different from the above values, proper allowance shall be made for the change in instrument reading.

* 4.6 Test methods.

* 4.6.1 Examination of product. Sample candles shall be examined to determine compliance with the workmanship, weight, and dimensions specified herein.

* 4.6.2 Effect of temperature. The candle shall be placed with axis horizontal (can lying on side) in a constant temperature cabinet and maintained at plus 125°F, ±3°F, for not less than 24 hours and the specimen observed periodically. The candle shall not show deformation. Wax flow shall be cause for rejection.

* 4.6.3 Burning quality. The flame shall consume the candle in not less than 10 hours under conditions free from drafts and at an ambient temperature of plus 70°F, ±5°F, without dripping, excessive smoking, or objectionable odor and without the necessity of trimming the wick.

5. PACKAGING

* 5.1 Preservation and packaging.

* 5.1.1 Level A. Candles shall be preserved in accordance with MIL-P-116, Method III and packaged in a domestic unit container conforming to PPP-B-676 or PPP-B-636 unless otherwise specified. A unit package shall contain 12 candles. The candles shall be separated by means of a sheet of paper having a minimum

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must have been manufactured within 6 months of the

weight of 25 pounds (24 by 36-500) per ream interleaved between the candles so that each candle is completely separated from the other candle and the surface of the container.

* 5.1.2 Level C. Candles shall be preserved and packaged in accordance with the manufacturer's commercial practice.

* 5.2 Intermediate packaging. Only identical items shall be included in an intermediate package. For items preserved and packaged to meet 5.1.2, requirements for intermediate packaging shall be at the option of the contractor. Candles preserved and packaged to meet 5.1.1 shall be packaged in domestic containers conforming to PPP-B-636. Unless otherwise specified by the procuring activity, the quantity of unit packages to be included in each intermediate package shall be at the option of the contractor and as governed by the weight limitations of the container being used.

5.3 Packing.

* 5.3.1 Level A. Candles preserved and packaged to meet 5.1.1 and 5.2 shall be packed in exterior type shipping containers, conforming to PPP-B-636. As far as practical, exterior containers shall be of uniform shape and size, be of minimum cube and tare consistent with the protection required and contain identical quantities. Closure of the containers shall be in accordance with the appendix of the box specification.

* 5.3.2 Level B. Candles preserved and packaged to meet 5.1.1 and 5.2 shall be packed in domestic type exterior containers conforming to PPP-B-636. Exterior containers shall be of minimum cube and tare consistent with the protection required. As far as practical, exterior containers shall be of uniform shape and size and contain identical quantities. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto.

* 5.3.3 Level C. Packages which require overpacking for acceptance by the Carrier shall be packed in exterior type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. Containers shall meet consolidated freight classification rules, or regulations of other Common Carriers as applicable to the mode of transportation.

* 5.4 Physical protection. Cushioning, blocking and bracing shall be in accordance with Method III of MIL-P-116.

5.5 Marking. Interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Candles covered by this specification are intended for use under emergency survival conditions.

6.2 Order data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Level of packaging and packing desired.

c. Preproduction requirements.

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* 6.3 Source of material. The aluminum can specified in 3.6 may be obtained from:

Allisco, Inc
4010 N. American Street
Philadelphia PA 19140

* 6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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☐ YES ☐ NO *(If "Yes", in what way?)*

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